

Software Requirements & Verification Document (SRVD)

1. DOCUMENT OVERVIEW

Document ID: SRVD-001

System Name: Autonomous Navigation Control System (ANCS)

Software Level: DAL-B (Design Assurance Level B)

Applicable Standard: DO-178C

Version: 1.0

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Purpose:

This document defines the software requirements, system architecture, verification strategy, and traceability framework for the Autonomous Navigation Control System (ANCS). The objective is to ensure compliance with DO-178C guidelines for safety-critical avionics software.

2. SYSTEM DESCRIPTION

The Autonomous Navigation Control System (ANCS) is responsible for real-time navigation and control decisions in an embedded avionics environment within an aircraft system.

The system operates as part of an aircraft navigation and control framework, ensuring safe and reliable behavior during all phases of aircraft operation, including normal and failure conditions.

The system performs the following functions:

- Real-time computation of navigation paths
- Processing and validation of sensor data inputs
- Continuous path correction and optimization
- Detection of faults and activation of fail-safe mechanisms

Operating Environment:

The system operates within a real-time embedded avionics platform using an RTOS-based architecture. The system must meet strict timing, safety, and reliability constraints.

Aircraft Context:

The ANCS operates within a safety-critical aircraft environment where software behavior directly impacts aircraft safety and operational reliability.

The system shall ensure predictable and safe operation of the aircraft under both nominal and failure conditions.

All navigation, control, and fault-handling mechanisms shall be designed to support aircraft-level safety objectives and regulatory compliance requirements.

3. SOFTWARE REQUIREMENTS

3.1 High-Level Requirements (HLR)

HLR-001:

The system shall compute the optimal navigation path at a fixed interval of 100 milliseconds to ensure timely decision-making in dynamic environments.

HLR-002:

The system shall detect sensor failure conditions within 200 milliseconds to prevent propagation of incorrect data.

HLR-003:

The system shall transition to a fail-safe operational mode upon detection of a critical system failure to ensure safety.

These high-level requirements define the functional behavior of the system and are derived from system-level safety and operational requirements.

3.2 Low-Level Requirements (LLR)

LLR-001:

The navigation engine shall implement the A* algorithm to compute optimal paths efficiently and deterministically.

LLR-002:

The system shall continuously monitor sensor heartbeat signals to detect anomalies or failures in sensor inputs.

LLR-003:

Upon detection of a critical failure, the system shall trigger a fallback control module to maintain safe operation.

These low-level requirements refine the high-level requirements and provide implementation-level details necessary for design and coding.

4. SOFTWARE ARCHITECTURE

The system architecture follows a modular layered design to ensure separation of concerns, maintainability, and fault isolation.

The major components include:

- Sensor Module: Responsible for acquiring and validating sensor data**
- Navigation Engine: Performs path computation and optimization**
- Control Unit: Executes navigation decisions**
- Fault Handler: Detects faults and manages fail-safe operations**

The architecture ensures:

- Loose coupling between components**
- Clearly defined interfaces**
- Isolation of faults to prevent system-wide failure**

Traceability between requirements and architecture components is maintained to ensure that all requirements are implemented.

5. VERIFICATION STRATEGY

5.1 Verification Methods

Verification activities include:

- Reviews:**

Systematic review of requirements and design to ensure correctness and completeness

- Analysis:**

Static analysis and traceability verification to confirm requirement consistency and coverage

- Testing:**

Execution of unit, integration, and system-level tests to validate functional behavior

5.2 Test Cases

TC-001:

Validates that the navigation path computation meets the timing constraint of 100 milliseconds.

TC-002:

Simulates sensor failure conditions and verifies that the system detects failures within 200 milliseconds.

TC-003:

Triggers a system fault and verifies that the system transitions to fail-safe mode.

Each test case is derived directly from high-level requirements, ensuring requirement-based testing.

6. TRACEABILITY

Traceability is maintained across all development stages.

The following relationships are established:

- High-Level Requirements → Low-Level Requirements**
- Low-Level Requirements → Implementation logic**
- Requirements → Test Cases**

Example mappings:

HLR-001 → LLR-001 → TC-001

HLR-002 → LLR-002 → TC-002

HLR-003 → LLR-003 → TC-003

This ensures bidirectional traceability, allowing verification of requirement coverage and validation of implementation correctness.

7. CONFIGURATION MANAGEMENT

All software artifacts are maintained under configuration control.

Key practices include:

- Version control using Git**
- Baseline definition for each release**
- Unique identification of all artifacts**
- Change tracking and auditability**

These practices ensure reproducibility and controlled evolution of the software system.

8. QUALITY ASSURANCE

An independent Software Quality Assurance (SQA) team is responsible for ensuring compliance with DO-178C objectives.

Activities include:

- Compliance checks against defined standards**
- Process audits and reviews**
- Verification of adherence to development and verification plans**

9. SAFETY CONSIDERATIONS

Safety is ensured through:

- Implementation of fail-safe mechanisms**
- Redundancy in sensor inputs to improve reliability**
- Continuous monitoring and error detection**
- Logging and recovery procedures for fault conditions**

The system ensures predictable and safe behavior under failure scenarios.

10. CONCLUSION

The ANCS software system is designed and verified in accordance with DO-178C guidelines.

The system demonstrates:

- Requirement traceability**
- Verification through testing and analysis**
- Compliance with safety and quality standards**

This ensures that the system meets the reliability and safety requirements expected of avionics software.

END OF DOCUMENT
